COS1501

UNIT 2

Rational and Real Numbers

$\mathbb{Q}$ • IRRATIONALS • $\mathbb{R}$

Complete, guide-aligned notes at the original academic level

Every assessed topic includes a medium and a really hard fully worked example

Colour key

TEAL	BLUE	PURPLE	RED / ORANGE
definitions and laws	medium worked examples	really hard worked examples	traps and exam notes

Unit roadmap and scope

These notes follow Study Unit 2 of the supplied COS1501 guide. They preserve its formal level and extend brief sections with complete algebra, proofs, counterexamples and exam-level applications.

EXAM NOTE - Required standard

You must be able to define each number system formally, perform exact rational arithmetic, identify illegal division by zero, construct and analyse contradiction proofs, and distinguish rational from irrational decimal behaviour.

PART	FOCUS	COMPLETE COVERAGE
1	Rational numbers	ratios p/q, $q\neq 0$, equivalent forms, arithmetic
2	Structure of $\mathbb{Q}$	identities, inverses, Law 11, linear equations
3	Geometry and parity	Pythagoras, even/odd/multiple forms
4	Proof and irrationality	reductio ad absurdum, proof that $\sqrt{2}\notin\mathbb{Q}$
5	Decimals and irrationals	repeating decimals, construction, approximation
6	Real numbers	$\mathbb{R}$, hierarchy, closure, square roots, quadratic formula

Key questions

- Why does $\mathbb{Z}$ need to be expanded to $\mathbb{Q}$, and why must every denominator be non-zero?
- Which laws survive in $\mathbb{Q}$, and which new multiplicative-inverse law becomes possible?
- How does a contradiction prove that $\sqrt{2}$ is irrational?
- How do repeating decimals characterise rationals and non-repeating decimals characterise irrationals?
- How do $\mathbb{Q}$ and the irrational numbers combine to form $\mathbb{R}$?

1. Why the integers must be extended

The equation $2x=1$ has no solution in $\mathbb{Z}$. Since $2\cdot0=0<1$ and $2\cdot1=2>1$, its solution lies strictly between 0 and 1. This forces an extension of the integer system.

DEFINITION - Number-system chain

$\mathbb{Z}^+ \subset \mathbb{Z}{\geq}0 \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Every extension solves equations that the earlier system cannot solve while retaining the earlier numbers.

The symbol $1/2$ denotes the number which, when multiplied by 2, gives 1. More generally, p/q denotes the number which, when multiplied by q, gives p - provided $q\neq0$.

MEDIUM WORKED EXAMPLE - An equation with no integer solution

Problem. Solve $7x=3$ and explain why $x\notin\mathbb{Z}$.

Solution.
Multiplying both sides by $1/7$ gives $x=3/7$. Since $0<3/7<1$ and no integer lies strictly between 0 and 1, $x\notin\mathbb{Z}$. But $x\in\mathbb{Q}$.

REALLY HARD WORKED EXAMPLE - Identify the smallest required system

Problem. For each equation, identify the smallest listed system $\mathbb{Z}^+$, $\mathbb{Z}{\geq}0$, $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$ containing its solutions: (a) $x+8=3$; (b) $6x=5$; (c) $x^2=2$.

Solution.
(a) $x=-5$, so $\mathbb{Z}$. (b) $x=5/6$, so $\mathbb{Q}$. (c) $x=\pm\sqrt{2}$, which are irrational real numbers, so $\mathbb{R}$. Each earlier system fails to contain at least one required value.

2. Rational numbers: formal definition and representation

DEFINITION - Formal definition

$\mathbb{Q} = \{p/q : p,q \in \mathbb{Z} \text{ and } q \neq 0\}$. A number is rational exactly when it can be represented as a ratio of two integers with a non-zero denominator.

A rational number has many representations: $2/3=4/6=-10/-15$. In lowest terms, numerator and denominator share no common factor greater than 1. A standard unique form also chooses a positive denominator.

Every integer n is rational because $n=n/1$. Thus $\mathbb{Z} \subset \mathbb{Q}$. A fraction bar is grouping: $a+b/c$ does not mean $(a+b)/c$ unless brackets or a long bar show that grouping.

MEDIUM WORKED EXAMPLE - Reduce to standard form

Problem. Write $-84/126$ in lowest terms with a positive denominator.

Solution.
$\gcd(84,126)=42$. Divide numerator and denominator by 42: $-84/126=-2/3$. The denominator is positive and $\gcd(2,3)=1$.

REALLY HARD WORKED EXAMPLE - Equality of rational representations

Problem. Prove that $a/b=c/d$, with $b,d\neq 0$, exactly when $ad=bc$.

Solution.
If $a/b=c/d$, multiply both sides by the non-zero number bd: $ad=bc$. Conversely, if $ad=bc$, multiply both sides by $1/(bd)$: $a/b=c/d$. The condition $b,d\neq 0$ is essential because the reciprocals and original ratios otherwise do not exist.

3. Why division by zero is impossible

The meaning p/q asks for a number x satisfying $qx=p$. Setting $q=0$ creates two fundamentally different failures.

DEFINITION - The two cases

If $p \neq 0$, $0x=p$ has no solution because $0x=0$ for every x.
If $p=0$, $0x=0$ has every number as a solution, so $0/0$ cannot denote one unique number.

Division must return a single well-defined number. Neither no answer nor infinitely many possible answers provides one. Therefore $p/0$ is undefined for every p, including $p=0$.

COMMON TRAP - Illegal cancellation

Cancelling a factor is multiplication by its reciprocal. You may cancel $x-a$ only after establishing $x-a \neq 0$. If $x=a$, cancellation would be division by zero.

MEDIUM WORKED EXAMPLE - Detect an undefined expression

Problem. Evaluate $(5-5)/(3-3)$ and explain the exact failure.

Solution.
The expression becomes $0/0$. It is not 0 and not 1; it is undefined because every x satisfies $0 \cdot x=0$, so no unique quotient is selected.

REALLY HARD WORKED EXAMPLE - Expose a fake proof

Problem. A calculation starts with $x=a$, factors $(x+a)(x-a)=a(x-a)$, cancels $x-a$, and concludes $2a=a$. Locate the invalid step.

Solution.
The premise $x=a$ makes $x-a=0$. Cancelling $x-a$ is therefore division by zero. Before cancellation both sides equal 0 and contain no information about $x+a$ versus a. The conclusion is unsupported.

4. Equivalent fractions and the hidden identity

Multiplying numerator and denominator by the same non-zero integer does not change a rational number:

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a/b · k/k = ak/bk, where k≠0 and k/k=1
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This uses the multiplicative identity. Cancelling is the reverse operation: $ak/bk=a/b$ when $k\neq0$. The restriction prevents an illegal zero denominator.

MEDIUM WORKED EXAMPLE - Create a required denominator

Problem. Express 7/12 with denominator 84.

Solution.
Since $84=12\cdot7$, multiply by 7/7: $7/12=49/84$.

REALLY HARD WORKED EXAMPLE - Recover an unknown equivalent fraction

Problem. Find all integers n for which $(3n-2)/(2n+5)=5/7$, subject to $2n+5\neq0$.

Solution.
Cross-multiply: $7(3n-2)=5(2n+5)$. Thus $21n-14=10n+25$, so $11n=39$. Hence $n=39/11$, which is not an integer. Therefore there is no integer n satisfying the equation. The denominator restriction is not the cause; integrality is.

5. Addition and subtraction in $\mathbb{Q}$

DEFINITION - General rules

$a/b + c/d = (ad+bc)/(bd)$, and $a/b - c/d = (ad-bc)/(bd)$, where $b,d \neq 0$.

A common denominator creates like-sized parts. Multiplying denominators always works, although the least common denominator often reduces arithmetic. Simplify only after the full numerator has been formed.

MEDIUM WORKED EXAMPLE - Add unlike rational numbers

Problem. Compute $7/18+5/24$ in lowest terms.

Solution.
$\mathrm{lcm}(18,24)=72$. Then $7/18=28/72$ and $5/24=15/72$. Sum: $43/72$. Since 43 is prime and does not divide 72, the fraction is reduced.

REALLY HARD WORKED EXAMPLE - A nested rational expression

Problem. Evaluate $5/6-[3/4-(7/10+1/15)]$ exactly.

Solution.
First $7/10+1/15=21/30+2/30=23/30$. Then $3/4-23/30=45/60-46/60=-1/60$. Finally $5/6-(-1/60)=50/60+1/60=51/60=17/20$.

6. Multiplication and division in $\mathbb{Q}$

DEFINITION - General rules

(a/b)(c/d)=ac/bd. Division by a non-zero rational c/d means multiplication by its reciprocal d/c:
(a/b) ÷ (c/d) = (a/b)(d/c), where b,c,d≠0.

Cross-cancellation before multiplication is legitimate when cancelling common non-zero factors. Never flip the first fraction when dividing; only the divisor is replaced by its reciprocal.

MEDIUM WORKED EXAMPLE - Multiply with cancellation

Problem. Compute (-14/15)(25/21) in lowest terms.

Solution.
Cancel 14 with 21 by 7, giving 2 and 3; cancel 25 with 15 by 5, giving 5 and 3. The result is -(2·5)/(3·3)=-10/9.

REALLY HARD WORKED EXAMPLE - Divide a product of ratios

Problem. Evaluate [(9/14)(-35/12)] ÷ (-15/8).

Solution.
Replace division by multiplication: (9/14)(-35/12)(-8/15). Two negatives give a positive. Cancel systematically: 35/14=5/2, 9/15=3/5, and 8/12=2/3. The remaining product is (3·5·2)/(2·3·5)=1.

7. Identities, additive inverses and multiplicative inverses in $\mathbb{Q}$

The additive identity 0 satisfies $x+0=x$. The multiplicative identity 1 satisfies $x \cdot 1=x$. Every rational x has an additive inverse $-x$. Every non-zero rational x has a multiplicative inverse $1/x$ satisfying $x(1/x)=1$.

DEFINITION - Reciprocal of a ratio

If $x=p/q\neq 0$, then $p\neq 0$ and $1/x=q/p$. Turning a fraction upside down is valid only when its numerator is non-zero.

Zero has no multiplicative inverse: a candidate y would require $0y=1$, contradicting $0y=0$.

MEDIUM WORKED EXAMPLE - Find both inverses

Problem. For $x=-7/12$, find its additive and multiplicative inverses and verify each.

Solution.
Additive inverse: $7/12$, since $-7/12+7/12=0$. Multiplicative inverse: $-12/7$, since $(-7/12)(-12/7)=1$.

REALLY HARD WORKED EXAMPLE - Inverse of a rational expression

Problem. For $x=(a-b)/(a+b)$, state all restrictions and find $1/x$.

Solution.
The original denominator requires $a+b\neq 0$. A multiplicative inverse additionally requires $x\neq 0$, so $a-b\neq 0$. Under both conditions, $1/x=(a+b)/(a-b)$.

8. Solving linear equations with multiplicative inverses

For $ax=b$ with $a \neq 0$, multiply both sides by $1/a$:

$$(1/a)(ax)=(1/a)b \rightarrow [(1/a)a]x=b/a \rightarrow 1x=b/a \rightarrow x=b/a$$

This is the precise algebra behind 'divide both sides by a'. If $a=0$, the equation has either no solution ($b \neq 0$) or every number as a solution ($b=0$).

MEDIUM WORKED EXAMPLE - Solve a rational linear equation

Problem. Solve $(7/9)x=-14/15$.

Solution.
Multiply by $9/7$: $x=(-14/15)(9/7)$. Cancel $14/7=2$ and $9/15=3/5$, obtaining $x=-6/5$.

REALLY HARD WORKED EXAMPLE - Parameter cases

Problem. Solve $ax=a+1$ over $\mathbb{Q}$, accounting for every possible rational a.

Solution.
If $a \neq 0$, multiply by $1/a$: $x=(a+1)/a=1+1/a$. If $a=0$, the equation becomes $0=1$, which has no solution. There is no 'all x' case because the right side is 1 when $a=0$.

9. The laws of $\mathbb{Q}$ and Law 11

$\mathbb{Q}$ retains the familiar laws: closure under + and ×; commutativity and associativity of both; distributivity; additive and multiplicative identities; additive inverses; no zero-divisors; linear order, monotonicity and transitivity.

DEFINITION - Law 11 - existence of multiplicative inverses

For every non-zero rational x, there exists a rational y such that $xy=1$. This law fails in $\mathbb{Z}$: for example, 2 has no integer multiplicative inverse because $2y=1$ has no integer solution.

MEDIUM WORKED EXAMPLE - Why $\mathbb{Z}$ fails Law 11

Problem. Use $x=5$ to test Law 11 in $\mathbb{Z}$ and $\mathbb{Q}$.

Solution.
In $\mathbb{Q}$, $y=1/5$ and $5(1/5)=1$. In $\mathbb{Z}$, no integer y satisfies $5y=1$. Thus the universal multiplicative-inverse law holds in $\mathbb{Q}$ but fails in $\mathbb{Z}$.

REALLY HARD WORKED EXAMPLE - Use laws to prove cancellation

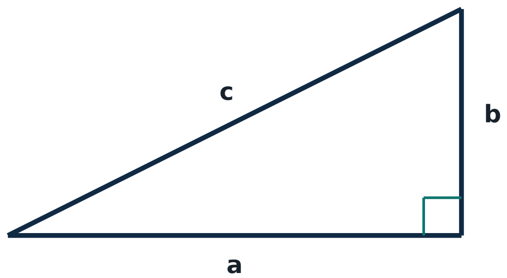
Problem. For $a,b,c\in\mathbb{Q}$ with $a\neq 0$, prove that $ab=ac$ implies $b=c$.

Solution.
Multiply both sides by $1/a$: $(1/a)(ab)=(1/a)(ac)$. By associativity, $[(1/a)a]b=[(1/a)a]c$. By Law 11, $(1/a)a=1$; by multiplicative identity, $1b=1c$, hence $b=c$. The condition $a\neq 0$ is essential.

10. The Theorem of Pythagoras

DEFINITION - Theorem

In a right-angled triangle, if c is the hypotenuse and a,b are the perpendicular legs, then $c^2=a^2+b^2$. The hypotenuse is opposite the right angle and is the longest side.



The theorem can find a missing side, but the computed square root may be irrational. Geometry therefore forces the extension from $\mathbb{Q}$ to $\mathbb{R}$.

MEDIUM WORKED EXAMPLE - Find a rational hypotenuse

Problem. A right triangle has legs 9 and 12. Find its hypotenuse.

Solution.
$c^2=9^2+12^2=81+144=225$, so $c=\sqrt{225}=15$. Length is non-negative, so the negative square root is rejected geometrically.

REALLY HARD WORKED EXAMPLE - Compare rational and irrational distances

Problem. From A=(-4,7), find the exact distances to B=(8,-2) and C=(9,-2), then classify both.

Solution.
For AB, the changes are 12 and -9: $AB^2=12^2+(-9)^2=225$, so AB=15, rational. For AC, the changes are 13 and -9: $AC^2=13^2+(-9)^2=250$, so $AC=\sqrt{250}=5\sqrt{10}$, irrational. Exact classification depends on whether the squared distance is a perfect square.

11. Even, odd and multiple forms

DEFINITION - General forms

An even integer has form $2k$. An odd integer has form $2k+1$, where $k\in\mathbb{Z}$. More generally, a multiple of n has form nk. Integers with remainder r on division by n have form $nk+r$.

These forms turn verbal parity claims into algebra. After simplification, showing an expression equals $2m$ proves it even; showing it equals $2m+1$ proves it odd.

MEDIUM WORKED EXAMPLE - Sum of even integers

Problem. Prove that the sum of two even integers is even.

Solution.
Let $m=2a$ and $n=2b$ for integers a,b. Then $m+n=2a+2b=2(a+b)$. Since $a+b\in\mathbb{Z}$, the sum has form $2k$ and is even.

REALLY HARD WORKED EXAMPLE - Square classification modulo four

Problem. Prove that every odd square has the form $4t+1$.

Solution.
Let $n=2k+1$. Then $n^2=4k^2+4k+1=4(k^2+k)+1$. Since $t=k^2+k\in\mathbb{Z}$, $n^2=4t+1$. Thus an integer square can never leave remainder 3 when divided by 4.

12. Proof by contradiction (reductio ad absurdum)

To prove a statement S by contradiction, assume its negation ¬S. Use valid definitions and laws until the assumption forces an impossibility. Since valid reasoning cannot produce a contradiction from true premises, ¬S must be false and S true.

DEFINITION - Proof skeleton

1. State the claim.
2. Assume the opposite.
3. Translate the assumption into precise algebra.
4. Derive a contradiction with a definition, theorem or earlier assumption.
5. Reject the opposite assumption and conclude the claim.

A contradiction is not merely a surprising result. It must be logically impossible, such as an integer being both odd and even, or a lowest-terms fraction having a common factor.

MEDIUM WORKED EXAMPLE - No largest integer

Problem. Prove by contradiction that there is no largest integer.

Solution.
Assume L is the largest integer. Then $L+1$ is an integer and $L+1>L$, contradicting that L is largest.
Therefore no largest integer exists.

REALLY HARD WORKED EXAMPLE - Infinitely many primes - Euclid's structure

Problem. Prove that no finite list can contain all primes.

Solution.
Assume $p_1,\ldots,p_n$ is the complete list. Form $N=p_1p_2\ldots p_n+1$. Dividing N by any listed prime leaves remainder 1, so none divides N. But $N>1$ has a prime divisor, which must be absent from the supposedly complete list. Contradiction; there are infinitely many primes.

13. The proof that $\sqrt{2}$ is irrational

The claim is that no rational number has square 2. The proof uses lowest terms, parity, and contradiction.

DEFINITION - Full proof

Assume $\sqrt{2}=p/q$ with $p,q\in\mathbb{Z}$, $q\neq 0$, and $\gcd(p,q)=1$. Squaring gives $p^2=2q^2$, so p^2 is even. An odd square is odd; therefore p is even, say $p=2k$. Substitute: $4k^2=2q^2$, so $q^2=2k^2$ and q is even. Thus p and q share factor 2, contradicting $\gcd(p,q)=1$. Hence $\sqrt{2}\notin\mathbb{Q}$.

The hidden lemma 'if p^2 is even, then p is even' is justified by its contrapositive: if p is odd, p^2 is odd. Skipping this bridge leaves a proof gap.

MEDIUM WORKED EXAMPLE - Locate the contradiction

Problem. Why is 'p and q are both even' a contradiction rather than merely another fact?

Solution.
The representation p/q was selected in lowest terms, so $\gcd(p,q)=1$. If both are even, 2 divides both, giving $\gcd(p,q)\geq 2$. This directly contradicts the lowest-terms condition.

REALLY HARD WORKED EXAMPLE - Adapt the method to $\sqrt{3}$

Problem. Prove $\sqrt{3}$ is irrational, using the fact that if 3 divides p^2 then 3 divides p.

Solution.
Assume $\sqrt{3}=p/q$ in lowest terms. Then $p^2=3q^2$, so $3|p^2$ and hence $3|p$; write $p=3k$. Substitution gives $9k^2=3q^2$, hence $q^2=3k^2$, so $3|q$. Then p and q share factor 3, contradicting lowest terms. Therefore $\sqrt{3}\notin\mathbb{Q}$.

14. Irrational numbers

DEFINITION - Definition

A real number is irrational when it is not rational - that is, it cannot be expressed as p/q with integers p,q and $q \neq 0$. Examples include $\sqrt{2}$, $\sqrt{3}$ and π.

Irrational does not mean imaginary, undefined or inaccurate. An irrational number is an exact real number with a definite position on the number line; only its decimal expansion is non-terminating and non-repeating.

MEDIUM WORKED EXAMPLE - Classify exact expressions

Problem. Classify $\sqrt{49}$, $\sqrt{50}$ and 0.125 as rational or irrational.

Solution.
$\sqrt{49}=7\in\mathbb{Q}$. $\sqrt{50}=5\sqrt{2}$ is irrational. $0.125=125/1000=1/8\in\mathbb{Q}$.

REALLY HARD WORKED EXAMPLE - When is a square root rational?

Problem. For positive integer n, explain why $\sqrt{n}$ is rational exactly when n is a perfect square.

Solution.
If $n=k^2$, then $\sqrt{n}=k\in\mathbb{Z}\subset\mathbb{Q}$. Conversely, suppose $\sqrt{n}=p/q$ in lowest terms. Then $p^2=nq^2$. Prime factorisation gives even exponents on p^2 and q^2, forcing every exponent in n to be even; hence n is a perfect square. Therefore a non-square integer has an irrational square root.

15. Arithmetic involving rational and irrational numbers

DEFINITION - Guaranteed results

If $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$, then $r+x$ is irrational. If $r \in \mathbb{Q}$, $r \neq 0$ and $x \notin \mathbb{Q}$, then rx is irrational. The non-zero condition on the product is essential.

Two irrational numbers may have a rational or irrational sum/product. Examples: $\sqrt{2}+(-\sqrt{2})=0$ rational, but $\sqrt{2}+\sqrt{3}$ irrational; $\sqrt{2}\cdot\sqrt{2}=2$ rational, but $\sqrt{2}\cdot\sqrt{3}=\sqrt{6}$ irrational.

MEDIUM WORKED EXAMPLE - Prove a rational plus irrational is irrational

Problem. Let r be rational and x irrational. Prove $r+x$ is irrational.

Solution.
Assume $r+x$ were rational. Since rational numbers are closed under subtraction, $(r+x)-r=x$ would be rational, contradicting that x is irrational. Hence $r+x$ is irrational.

REALLY HARD WORKED EXAMPLE - Prove the product rule and find the exception

Problem. Let $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$. Prove rx is irrational when $r \neq 0$, then show why the condition cannot be removed.

Solution.
Assume $rx \in \mathbb{Q}$. Because $r \neq 0$, $1/r \in \mathbb{Q}$. Then $x=(1/r)(rx)$ would be rational by closure of $\mathbb{Q}$ under multiplication - contradiction. If $r=0$, $rx=0$ for every x, and 0 is rational. Thus $r \neq 0$ is necessary.

16. Decimal expansions of rational numbers

DEFINITION - Characterisation

A real number is rational exactly when its decimal expansion terminates or eventually repeats. A terminating decimal can be viewed as repeating zeros.

Long division has only finitely many possible remainders. Once a remainder repeats, the later digits repeat in a cycle. If remainder 0 occurs, the expansion terminates.

MEDIUM WORKED EXAMPLE - Convert a repeating decimal to a ratio

Problem. Write 0.777... (the digit 7 repeats forever) as a fraction.

Solution.
Let $x=0.777...$. Then $10x=7.777...$. Subtract: $10x-x=7$, so $9x=7$ and $x=7/9$.

REALLY HARD WORKED EXAMPLE - Convert a mixed repeating decimal

Problem. Convert 2.13272727... (the block 27 repeats forever) to a fraction.

Solution.
Let $x=2.13272727...$. The non-repeating part has two decimal places and the repeating block has two. Then $10000x=21327.2727...$ and $100x=213.2727...$. Subtract: $9900x=21114$, so $x=21114/9900=3519/1650=1173/550$.

NOTE - Notation

Because overbars can render ambiguously, always state the repeating block in words or brackets, for example 0.13(27)(27)... .

17. Non-terminating, non-repeating decimals

An irrational decimal continues forever without eventually repeating a fixed finite block. 'Non-terminating' alone is insufficient: 1/3=0.333... is non-terminating but rational because it repeats.

A deliberate pattern such as 0.10110111011110... can be designed so runs change length forever, preventing eventual periodicity. The written digits specify an exact real number, not an approximation.

MEDIUM WORKED EXAMPLE - Classify by decimal behaviour

Problem. Classify 0.121212..., 0.25000... and 0.101001000100001... .

Solution.
0.121212... repeats '12', so rational. 0.25000... terminates, so rational. In 0.101001000100001..., the gaps of zeros grow, so no fixed block repeats eventually; it is irrational.

REALLY HARD WORKED EXAMPLE - Disprove a bad criterion

Problem. A student says: 'Every infinite decimal is irrational.' Give a counterexample and replace the claim with a correct biconditional.

Solution.
Counterexample: 0.333...=1/3 is infinite but rational. Correct statement: a real number is rational iff its decimal expansion terminates or eventually repeats; it is irrational iff the decimal is non-terminating and non-eventually-repeating.

18. Approximating irrational numbers

Although $\sqrt{2}$ has no terminating exact decimal, it can be bracketed to any required precision. Because squaring is increasing for positive numbers, compare candidate squares with 2.

$$1^2<2<2^2 \rightarrow 1<\sqrt{2}<2 \quad 1.4^2<2<1.5^2 \rightarrow 1.4<\sqrt{2}<1.5$$

Refine one decimal place at a time. A calculator display is a rational approximation, not the exact irrational number.

MEDIUM WORKED EXAMPLE - Bracket $\sqrt{2}$ to three decimals

Problem. Show that $1.414<\sqrt{2}<1.415$.

Solution.
$1.414^2=1.999396<2$, while $1.415^2=2.002225>2$. Since both endpoints are positive, taking square-root order gives $1.414<\sqrt{2}<1.415$.

REALLY HARD WORKED EXAMPLE - Certify a rounded value

Problem. Prove that $\sqrt{7}$ rounds to 2.646 to three decimal places.

Solution.
For rounding to 3 decimals, show $2.6455\leq\sqrt{7}<2.6465$. Squaring gives $2.6455^2=6.99867025<7$ and $2.6465^2=7.00396225>7$. Therefore $\sqrt{7}$ lies in the rounding interval and rounds to 2.646.

19. Real numbers: $\mathbb{R}$

DEFINITION - Definition

$\mathbb{R}$ is the set consisting of all rational and irrational numbers. Symbolically, $\mathbb{R}=\mathbb{Q}\cup(\mathbb{R}\setminus\mathbb{Q})$, and every point on the ordinary number line corresponds to a real number.

The hierarchy is $\mathbb{Z}^+\subset\mathbb{Z}_{\geq 0}\subset\mathbb{Z}\subset\mathbb{Q}\subset\mathbb{R}$. Real addition and multiplication obey the familiar laws, but a rigorous construction of arithmetic on infinite decimals requires limits, beyond Unit 2. Complex numbers lie outside this unit.

MEDIUM WORKED EXAMPLE - Classify within the hierarchy

Problem. Place -3, 4/9, $\sqrt{5}$ and $\sqrt{36}$ in the smallest standard set among $\mathbb{Z}^+$, $\mathbb{Z}_{\geq 0}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.

Solution.
-3: $\mathbb{Z}$. 4/9: $\mathbb{Q}$. $\sqrt{5}$: $\mathbb{R}$ but not $\mathbb{Q}$. $\sqrt{36}=6$: $\mathbb{Z}^+$.

REALLY HARD WORKED EXAMPLE - Analyse an expression without decimals

Problem. Classify $(\sqrt{18}-\sqrt{8})/\sqrt{2}$.

Solution.
$\sqrt{18}=3\sqrt{2}$ and $\sqrt{8}=2\sqrt{2}$. Thus $(\sqrt{18}-\sqrt{8})/\sqrt{2}=(3\sqrt{2}-2\sqrt{2})/\sqrt{2}=\sqrt{2}/\sqrt{2}=1$. Although the expression contains irrational components, its exact value is the positive integer 1.

20. Expressing zero as a ratio

Zero is rational because $0=0/b$ for every non-zero integer b. The denominator cannot be zero, so $0/0$ is excluded.

DEFINITION - Why zero has no reciprocal

A multiplicative inverse y of 0 would satisfy $0y=1$. But multiplication by zero gives $0y=0$ for every y. Therefore no multiplicative inverse of zero exists in $\mathbb{Q}$ or $\mathbb{R}$.

MEDIUM WORKED EXAMPLE - Valid zero ratios

Problem. Which of $0/7$, $0/(-11)$, $5/0$ and $0/0$ represent zero?

Solution.
$0/7=0$ and $0/(-11)=0$ because denominators are non-zero. Both $5/0$ and $0/0$ are undefined.

REALLY HARD WORKED EXAMPLE - Solve a parameter ratio

Problem. For integer k, determine when $(k^2-9)/(k-3)$ equals 0, and account for the denominator.

Solution.
A ratio equals 0 when its numerator is 0 and denominator non-zero. $k^2-9=(k-3)(k+3)=0$ gives $k=3$ or -3. At $k=3$ the denominator is 0, so the expression is undefined. At $k=-3$ the denominator is -6, so the value is 0. Only $k=-3$ works.

21. n/n=1, integers as fractions, and fraction terminology

For every non-zero number n, $n/n=1$ because 1 is the number that multiplies n to return n. The condition $n \neq 0$ matters: $0/0$ is undefined.

Every integer n can be written $n/1$, so every integer is a fraction in the ratio sense. A proper fraction usually has $|numerator|<|denominator|$; an improper fraction has numerator magnitude at least as large as denominator magnitude.

MEDIUM WORKED EXAMPLE - Interpret fraction language

Problem. Classify 7/9, -13/5 and 42 as rational numbers and as proper/improper ratios.

Solution.
7/9 is proper. -13/5 is improper by magnitude. 42=42/1 is an improper ratio. All three are rational.

REALLY HARD WORKED EXAMPLE - Separate value from representation

Problem. Are 6/6, (-6)/(-6), 0/6 and 0/0 equal? Explain without informal cancellation.

Solution.
6/6=1 and (-6)/(-6)=1 because denominators are non-zero and each asks what multiplies the denominator to recover the numerator. 0/6=0. The expression 0/0 is undefined, so it cannot be declared equal to any of the others.

22. The quadratic formula

The guide names the quadratic formula as a Unit 2 concept. For $ax^2+bx+c=0$ with $a\neq 0$:

DEFINITION - Quadratic formula

$x = [-b \pm \sqrt{(b^2-4ac)}]/(2a)$. The quantity $\Delta=b^2-4ac$ is the discriminant.

The formula comes from completing the square, not memorisation alone:

$$ax^2+bx+c=0$$
$$x^2+(b/a)x=-c/a$$
$$(x+b/(2a))^2=(b^2-4ac)/(4a^2)$$
$$x=[-b\pm\sqrt{(b^2-4ac)}]/(2a)$$

Over $\mathbb{R}$: $\Delta>0$ gives two distinct real roots; $\Delta=0$ gives one repeated real root; $\Delta<0$ gives no real roots. If Δ is a non-square positive integer, the real roots are irrational.

MEDIUM WORKED EXAMPLE - Two rational roots

Problem. Solve $3x^2-5x-2=0$.

Solution.
$a=3,b=-5,c=-2$. $\Delta=(-5)^2-4(3)(-2)=25+24=49$. Therefore $x=[5\pm 7]/6$, giving $x=2$ or $x=-1/3$.

REALLY HARD WORKED EXAMPLE - Irrational roots and classification

Problem. Solve $2x^2+4x-7=0$ exactly and classify the roots.

Solution.
$a=2,b=4,c=-7$. $\Delta=16-4(2)(-7)=72$. $\sqrt{72}=6\sqrt{2}$. Thus $x=[-4\pm 6\sqrt{2}]/4=[-2\pm 3\sqrt{2}]/2$. Since a non-zero rational plus a non-zero rational multiple of $\sqrt{2}$ is irrational, both roots are irrational real numbers.

COMMON TRAP - Formula trap

The denominator is $2a$ under the entire numerator $-b\pm\sqrt{\Delta}$. Do not divide only the radical by $2a$.

23. Master comparison matrix

This matrix consolidates the exact structural differences tested across Units 1 and 2.

Feature	$\mathbb{Z}$	$\mathbb{Q}$	Irrationals	$\mathbb{R}$
Example	-4	7/9	√2	all previous
Closed under +	Yes	Yes	No	Yes
Closed under ×	Yes	Yes	No	Yes
Every non-zero has reciprocal inside	No	Yes	Yes	Yes
Decimal behaviour	terminating	terminating/repeating	non-repeating	either
Contains √2	No	No	Yes	Yes
Contains all earlier systems	No	contains $\mathbb{Z}$	not a superset of $\mathbb{Q}$	contains $\mathbb{Q}$

MEDIUM WORKED EXAMPLE - Test irrational closure

Problem. Give two irrational numbers whose sum is rational.

Solution.
Take √2 and -√2. Both are irrational, but their sum is 0, which is rational. Therefore the irrationals are not closed under addition.

REALLY HARD WORKED EXAMPLE - Classify by structure

Problem. Classify [(√3+1)(√3-1)]/(5/2) in the smallest standard system.

Solution.
The numerator is a difference of squares: 3-1=2. Then 2÷(5/2)=2·(2/5)=4/5. This lies in $\mathbb{Q}$ but not $\mathbb{Z}$, so $\mathbb{Q}$ is the smallest standard system.

24. Proof and counterexample workshop

The study guide's self-assessment requires substantiation, not unsupported yes/no answers. Use a general form for a proof; use one legal instance for a counterexample.

MEDIUM WORKED EXAMPLE - Even prime numbers

Problem. Prove that 2 is the only even prime.

Solution.
Let p be an even integer greater than 2. Then $p=2k$ with integer $k>1$, so p has factors 2 and k in addition to 1 and itself. Hence p is composite. The number 2 itself has only factors 1 and 2, so it is the unique even prime.

REALLY HARD WORKED EXAMPLE - Product of odd integers

Problem. Prove that the product of two odd integers is odd, and state exactly where closure is used.

Solution.
Let $m=2a+1$ and $n=2b+1$. Then $mn=4ab+2a+2b+1=2(2ab+a+b)+1$. Since $a,b\in\mathbb{Z}$ and $\mathbb{Z}$ is closed under addition and multiplication, $k=2ab+a+b$ is an integer. Thus $mn=2k+1$ and is odd.

EXAM NOTE - Counterexample to false distributivity

The claim $m+nk=(m+n)(m+k)$ for all positive integers is false. With $m=1,n=2,k=3$, left side=7 and right side=12.

25. Exam traps checklist

COMMON TRAP - Check

Never permit $q=0$ in p/q.

COMMON TRAP - Check

Do not assign any value to $0/0$; it is indeterminate/undefined, not automatically 0 or 1.

COMMON TRAP - Check

Do not cancel a factor that might be zero.

COMMON TRAP - Check

When adding fractions, do not add denominators.

COMMON TRAP - Check

A reciprocal exists only for a non-zero number.

COMMON TRAP - Check

A non-terminating decimal can still be rational if it eventually repeats.

COMMON TRAP - Check

Do not claim sums or products of two irrationals must be irrational.

COMMON TRAP - Check

In the $\sqrt{2}$ proof, justify why p^2 even forces p even.

COMMON TRAP - Check

A contradiction must conflict with a stated assumption or proven fact.

COMMON TRAP - Check

The calculator decimal for $\sqrt{2}$ is an approximation, not the exact value.

COMMON TRAP - Check

For Pythagoras, identify the hypotenuse before substituting.

COMMON TRAP - Check

In the quadratic formula, the whole numerator is divided by $2a$.

26. Final mixed challenge

Attempt these closed-book. Full worked answers follow so you can mark the method and the conclusion.

1. Write -18/42 in standard lowest-terms form.

2. Evaluate 5/12-7/18+11/24.

3. Explain separately why 8/0 and 0/0 are undefined.

4. Solve (9/14)x=27/35.

5. Prove the square of an even integer is divisible by 4.

6. Convert 0.181818... to a fraction.

7. Classify √81, √12, 0.272727... and π.

8. Find the exact hypotenuse for legs 7 and 11.

9. Prove that 4+√5 is irrational.

10. Solve x^2-6x+1=0 exactly and classify its roots.

Worked solutions

1. gcd(18,42)=6, so -18/42=-3/7.

2. Common denominator 72: 30/72-28/72+33/72=35/72.

3. 8/0 asks for x with 0x=8; none exists. 0/0 asks for x with 0x=0; every x works, so no unique quotient exists.

4. x=(27/35)(14/9)=6/5.

5. Let n=2k. Then $n^2=4k^2$, a multiple of 4.

6. Let x=0.181818.... Then 100x-x=18, so 99x=18 and x=2/11.

7. √81=9∈$\mathbb{Z}^+$; √12=2√3 is irrational; 0.272727...=3/11∈$\mathbb{Q}$; π is irrational real.

8. c=√(7^2+11^2)=√170, irrational because 170 is not a perfect square.

9. If 4+√5 were rational, subtracting rational 4 would make √5 rational, contradiction.

10. x=[6±√(36-4)]/2=[6±4√2]/2=3±2√2. Both roots are irrational real numbers.

27. One-page revision sheet

CONCEPT	MUST-KNOW FORM
Rational	p/q with $p,q \in \mathbb{Z}$ and $q \neq 0$
Division by zero	$p\neq0$: no solution; $0/0$: no unique solution
Add/subtract	$a/b \pm c/d = (ad \pm bc)/bd$
Multiply	$(a/b)(c/d)=ac/bd$
Divide	multiply by reciprocal; divisor must be non-zero
Law 11	each non-zero rational has multiplicative inverse
Pythagoras	$c^2=a^2+b^2$, c hypotenuse
Parity	even$=2k$; odd$=2k+1$
Contradiction	assume negation; derive impossibility; reject assumption
$\sqrt{2}$ proof	lowest terms → p even → q even → contradiction
Irrational	real but not p/q
Rational decimals	terminate or eventually repeat
Irrational decimals	non-terminating and non-eventually-repeating
Real	rational ∪ irrational
Quadratic formula	$x=[-b\pm\sqrt{(b^2-4ac)}]/(2a)$

NOTE - Source alignment

Coverage is based on Study Unit 2 (guide pages 21-32) of the supplied COS1501 study guide. The explanations and worked examples are newly written and deepen the guide without lowering its level.

EXAM NOTE - Readiness standard

Move on only when you can prove parity claims, explain both zero-denominator failures, reproduce the $\sqrt{2}$ contradiction proof, convert repeating decimals exactly, and classify expressions without relying on approximate decimals.